

Day 6: Introduction to Network DevicesTheory

Task1 :- Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

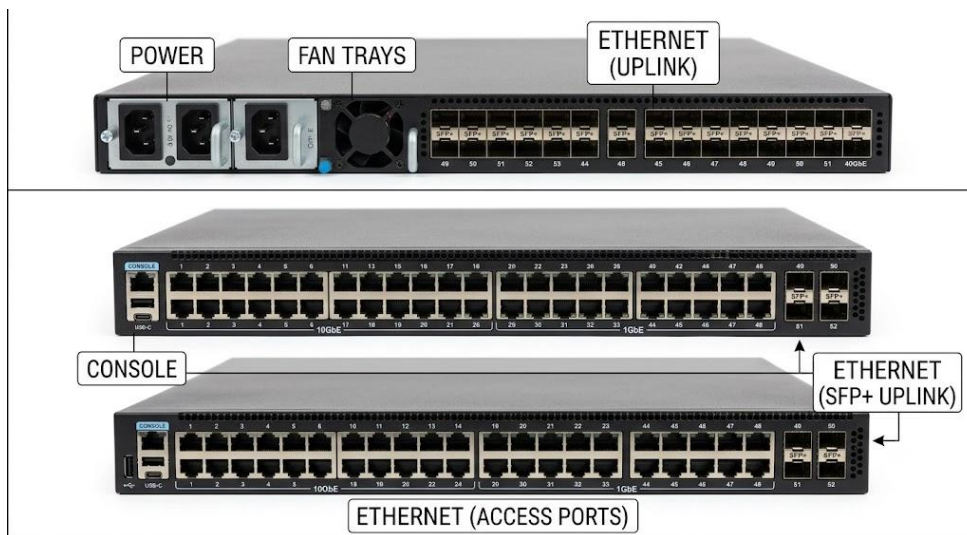
ANS :-

Device	Main Function	Typical Ports Used
Hub	Broadcasts all data to every connected device (outdated and simple).	LAN (RJ-45)
Switch	Connects multiple wired devices together on the same local network.	LAN (RJ-45)
Router	Connects your home network to the internet and directs network traffic.	WAN (to modem/internet), LAN (to wired devices)
Firewall	Blocks unauthorized access and filters dangerous internet traffic.	Built directly into the WAN/LAN ports on standard home routers.
Access Point (AP)	Broadcaster that sends out the wireless Wi-Fi signal to your devices.	LAN (RJ-45, often with PoE power)

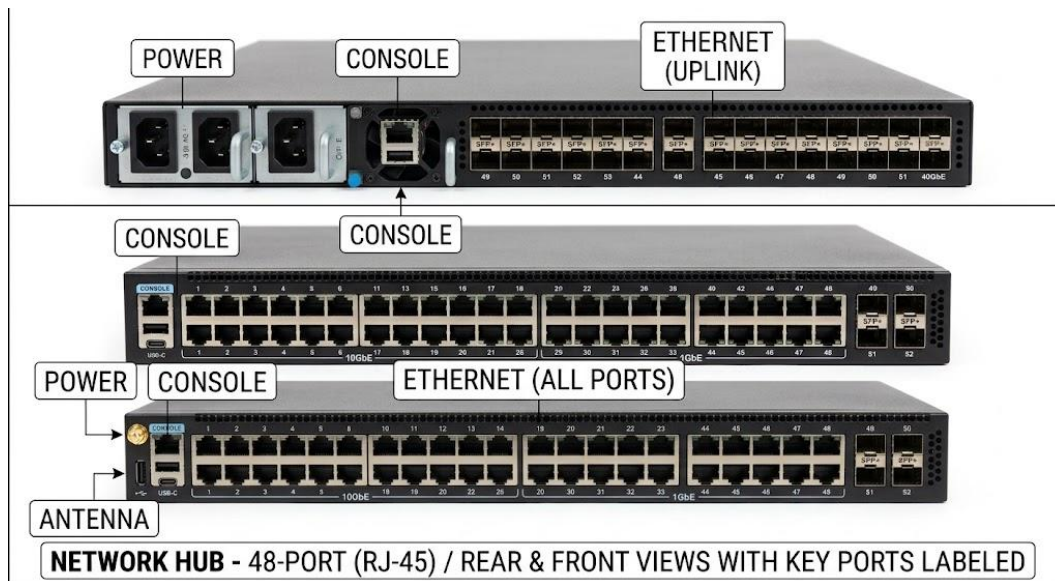
Task 2 :- Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.

ANS :-

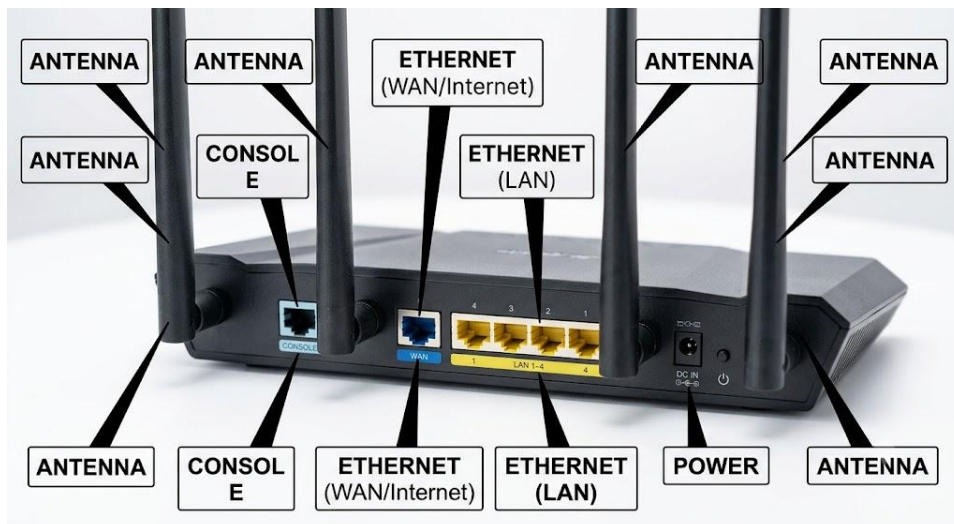
Switch :-



HUB :-



ROUTER :-



Task 3 :- Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.

ANS :-

Router :

Directs that request out of your local network and across the internet to Instagram's servers, then safely guides the returning photos right back to your phone.

Firewall :

Checks all incoming traffic to make sure it is safe data from Instagram, blocking any dangerous viruses or hackers before they reach your device.

Task 4 :- Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

ANS :-

Possible Reason :

The Access Point (AP) is overloaded. College Wi-Fi slows down when too many students connect to the same AP at once, causing signal drops and congestion.

The Solution :

Upgrade to a modern Wi-Fi 6 Access Point. A newer AP handles hundreds of simultaneous devices without slowing down, keeping your connection fast and stable.